

Fourier series of continuous functions

a calculation for weeks 4 & 5 (*version 2*)

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UAF Math 617 Functional Analysis

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Outline

- 1 sines (and complex exponentials) are orthonormal
- 2 what did Fourier believe? (1822)
- 3 what did Dirichlet prove? (1829)
- 4 what is the full story for Fourier series on $C[-\pi, \pi]$? (1966)
- 5 what is the clean L^2 version of the story? (~ 1910)
- 6 theory from the book

the product-of-sines integral

- for $n \in \mathbb{N}$, $\sin(nx)$ is a *wave* on $[0, \pi]$
- suppose $m, n \in \mathbb{N}$ and integrate the product of sines:

$$\int_0^\pi \sin(mx) \sin(nx) dx =$$

$$= \begin{cases} \frac{\pi}{2}, & m = n \\ 0, & \text{otherwise} \end{cases}$$

$$\cos(\theta \pm \psi) = \cos \theta \cos \psi \mp \sin \theta \sin \psi \quad \therefore \quad \sin \theta \sin \psi = \frac{1}{2} (\cos(\theta - \psi) - \cos(\theta + \psi))$$

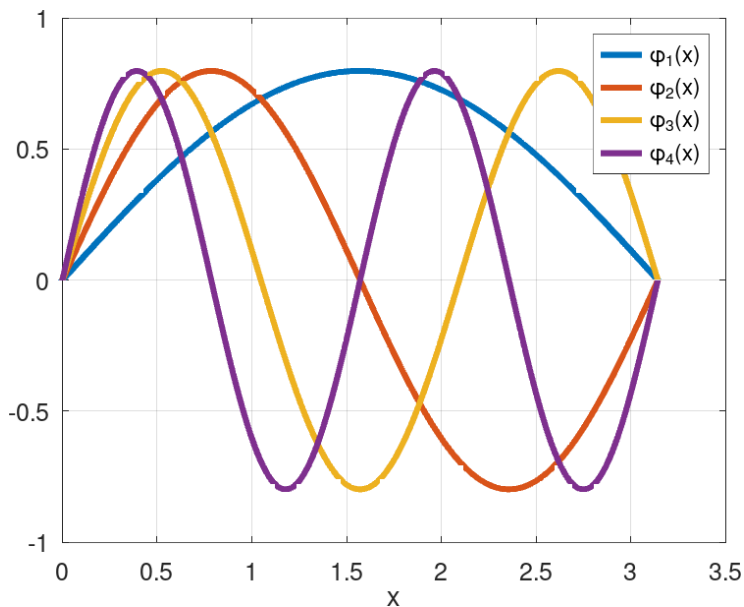
an infinite orthonormal set in $C[0, \pi]$

- fact from last slide: $\int_0^\pi \sin(mx) \sin(nx) dx = \frac{\pi}{2} \delta_{mn}$
- define

$$\phi_n(x) = \sqrt{\frac{2}{\pi}} \sin(nx) \quad \text{and} \quad S = \{\phi_n(x) : n \in \mathbb{N}\}$$

- then:
 - 1 S is orthonormal in the (real) inner product $\langle f, g \rangle = \int_0^\pi f(x)g(x) dx$:
 - 2 S is linearly independent:
- questions: what is $\text{span}(S)$? what is $\overline{\text{span}(S)}$?

an infinite orthonormal set in $C[0, \pi]$



a complex-valued orthonormal set in $C[-\pi, \pi]$

- easier integral:

$$\int_{-\pi}^{\pi} e^{imx} e^{-inx} dx = \int_{-\pi}^{\pi} e^{i(m-n)x} dx = 2\pi \delta_{mn}$$

- define

$$\psi_n(x) = \frac{1}{\sqrt{2\pi}} e^{inx} \quad \text{and} \quad E = \{\psi_n(x) : n \in \mathbb{Z}\}$$

- recall *complex* inner product (*sesquilinear* :-), now on $C[-\pi, \pi]$:

$$\langle f, g \rangle = \int_{-\pi}^{\pi} f(x) \overline{g(x)} dx$$

- then:

- 1 E is orthonormal
- 2 E is linearly-independent

- same big *question*: what is $\overline{\text{span}(E)}$?

before we move on ...

basic facts about $e^{i\theta}$:

- $e^{i\theta} = \cos \theta + i \sin \theta$
 - derive this from $e^z = 1 + z + \frac{z^2}{2!} + \cdots + \frac{z^n}{n!} + \cdots$
 - also recall Taylor series for $\cos$ and $\sin$
- $\cos \theta = \frac{1}{2} (e^{i\theta} + e^{-i\theta}), \quad \sin \theta = \frac{1}{2i} (e^{i\theta} - e^{-i\theta})$

basic facts about even and odd Fourier series:

- if f is even then

$$\int_{-\pi}^{\pi} f(x) e^{-inx} dx = \int_{-\pi}^{\pi} f(x) \cos(nx) dx = 2 \int_0^{\pi} f(x) \cos(nx) dx$$

- if f is odd then

$$\int_{-\pi}^{\pi} f(x) e^{-inx} dx = i \int_{-\pi}^{\pi} f(x) \sin(nx) dx = 2i \int_0^{\pi} f(x) \sin(nx) dx$$

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Fourier's assertion

claim (Fourier, *Théorie Analytique de la Chaleur*, 1822)

if $f \in C[0, \pi]$, and if we compute these coefficients

$$c_n = \langle f, \phi_n \rangle = \sqrt{\frac{2}{\pi}} \int_0^\pi f(x) \sin(nx) dx$$

for $n \in \mathbb{N}$, then

$$f(x) = \sum_{n=1}^{\infty} c_n \phi_n(x) = \sqrt{\frac{2}{\pi}} \sum_{n=1}^{\infty} c_n \sin(nx)$$

- the result is the *Fourier sine series* (or *expansion*) of f
- actually Fourier claimed this for discontinuous functions too
 - observe that c_n is an *integral*, so only features of f that change the integrals can affect the expansion
 - Fourier, and everybody else, agreed that the claim could not be exactly true at the points of discontinuity of $f(x)$

complex version is easier

- Fourier used real numbers (to my knowledge)
- we may replace

$$C[0, \pi] \rightarrow C[-\pi, \pi] \quad \text{and} \quad \sin(nx) \rightarrow e^{inx},$$

and use the complex (sesquilinear) inner product

Fourier's claim for complex exponentials

if $f \in C[-\pi, \pi]$, complex-valued, and if

$$c_n = \langle f, \psi_n \rangle = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

for $n \in \mathbb{Z}$, then

$$f(x) = \sum_{n \in \mathbb{Z}} c_n \psi_n(x) = \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} c_n e^{inx}$$

- this is the *Fourier series* of f , or the *complex Fourier series*
- for formulas from different references, be careful where the 2π goes

example 1: $f(x) = x$

- consider the Fourier sine series for $f(x) = x$, using the real orthonormal set $\{\phi_n(x) = \sqrt{\frac{2}{\pi}} \sin(nx)\} \subset C[0, \pi]$
- integrate to get coefficients:

$$c_n = \langle f, \phi_n \rangle = \sqrt{\frac{2}{\pi}} \int_0^\pi x \sin(nx) dx =$$

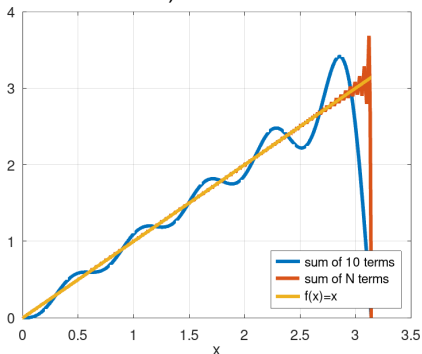
$$= \frac{\sqrt{2\pi}(-1)^{n-1}}{n}$$

example 1: $f(x) = x$

- MATLAB:

```
f = @(x) x;  
x = 0:pi/300:pi; N = 150; nn = 1:N;  
c = sqrt(2 * pi) * (-1).^(nn - 1) ./ nn;  
sN = zeros(size(x));  
for n=1:N  
    sN = sN + c(n) * sqrt(2/pi) * sin(n*x); end  
plot(x, f(x), x, sN)
```

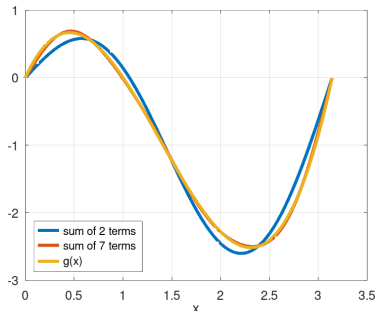
- result (works ...but *Gibbs effect*!):



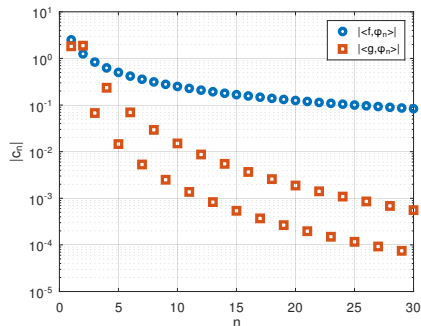
example 2: $g(x)$ cubic

- note that if $f(x) = x$ is *extended periodically* to the real line then the result is *not* continuous
- instead consider $g(x) = x(x - 1)(x - \pi)$ on the interval $[0, \pi]$
- now the periodic extension *is* continuous
- coefficients, done numerically: $c_n = \langle g, \phi_n \rangle = \sqrt{\frac{2}{\pi}} \int_0^\pi g(x) \sin(nx) dx$

MATLAB result:



coefficient decay:



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- again consider complex Fourier series for $f \in C[-\pi, \pi]$, namely

$$c_n = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} f(x) e^{-inx} dx, \quad f(x) = \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} c_n e^{inx}$$

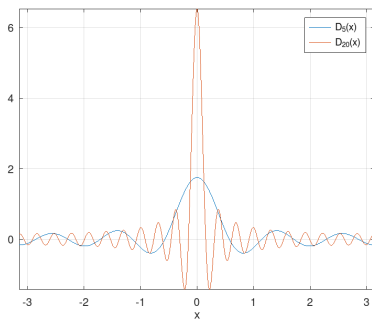
- combine into one formula and exchange limit processes:

$$\begin{aligned} f(x) &= \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} \left(\frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} f(y) e^{-iny} dy \right) e^{inx} \\ &= \int_{-\pi}^{\pi} \left(\frac{1}{2\pi} \sum_{n \in \mathbb{Z}} e^{-iny} e^{inx} \right) f(y) dy = \int_{-\pi}^{\pi} \left(\frac{1}{2\pi} \sum_{n \in \mathbb{Z}} e^{in(x-y)} \right) f(y) dy \end{aligned}$$

- exchange of integral and sum is justified if $\sum |c_n| < \infty$
- if the **inner sum** is approximated by a partial sum, then we are approximating f by integrating it against a kernel

Definition (Dirichlet's kernel, 1829)

$$D_m(x) = \frac{1}{2\pi} \sum_{n=-m}^m e^{inx} \stackrel{\text{why?}}{=} \frac{1}{2\pi} + \frac{1}{\pi} \sum_{n=1}^m \cos(nx)$$



Fourier's claim has become:

$$\begin{aligned} f(x) &= \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} c_n e^{inx} \\ &= \lim_{m \rightarrow \infty} \int_{-\pi}^{\pi} D_m(x-y) f(y) dy \end{aligned}$$

Definition (Dirichlet's kernel)

$$D_m(x) = \frac{1}{2\pi} \sum_{n=-m}^m e^{inx}$$

Lemma (properties of the kernel)

- ① D_m is continuous,
- ② $\int_{-\pi}^{\pi} D_m(x) dx = 1$,
- ③ $D_m(0) = \frac{m+1/2}{\pi}$, thus $D_m(0) \rightarrow \infty$ as $m \rightarrow \infty$, and
- ④ $D_m(x) = \frac{1}{2\pi} \frac{\sin((m+1/2)x)}{\sin(x/2)}$ for $x \neq 0$.

Proof. Properties ①, ②, ③ are easy from the definition of D_m . Property ④ is on the next slide. □

Dirichlet's kernel as a sine ratio

Property 4: $D_m(x) = \frac{1}{2\pi} \frac{\sin((m+1/2)x)}{\sin(x/2)}$

Proof. Apply knowledge of **geometric series**, then use $\sin \theta = (e^{i\theta} - e^{-i\theta})/(2i)$:

$$\begin{aligned}\sum_{n=-m}^m e^{inx} &= \sum_{n=-m}^m (e^{ix})^n = e^{-imx} \sum_{n=0}^{2m} (e^{ix})^n \\ &= e^{-imx} \frac{1 - (e^{ix})^{2m+1}}{1 - e^{ix}} = \frac{e^{-imx} - e^{i(m+1)x}}{1 - e^{ix}} \\ &= \frac{e^{-i(m+1/2)x} - e^{i(m+1/2)x}}{e^{-i(x/2)} - e^{i(x/2)}} \\ &= \frac{\sin((m+1/2)x)}{\sin(x/2)}\end{aligned}$$

The above calculation applies for $x \neq 0$.



Dirichlet's theorem (1829)

Theorem

If $f \in C[-\pi, \pi]$ is periodic, and if $f'(x)$ exists at $x \in [-\pi, \pi]$, then

$$\lim_{m \rightarrow \infty} \int_{-\pi}^{\pi} D_m(x-y) f(y) dy = f(x),$$

and thus Fourier's claim is true:

$$f(x) = \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} c_n e^{inx} \quad \text{for } c_n = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

- The proof is based on the *Riemann-Lebesgue lemma*. This version only requires the Riemann integral, and Dirichlet must have known it. I will prove it after using it.

Lemma (Riemann-Lebesgue for continuous functions)

$$\text{If } g \in C[a, b] \text{ then } \lim_{n \rightarrow \infty} \int_a^b g(x) e^{inx} dx = 0.$$

Dirichlet's theorem (1829)

Proof. By the periodicity of D_m and f , and since D_m is even, use $y = x + \xi$ to show

$$\int_{-\pi}^{\pi} D_m(x - y) f(y) dy = \int_{-\pi}^{\pi} D_m(\xi) f(x + \xi) d\xi$$

By replacing $f(x + \xi)$ with $\tilde{f}(\xi)$, we may assume $x = 0$ and $f'(0)$ exists. Next we calculate using the proven properties of the kernel:

$$\begin{aligned} \int_{-\pi}^{\pi} D_m(\xi) f(\xi) d\xi &= \int_{-\pi}^{\pi} D_m(\xi) (f(\xi) - f(0)) d\xi + \int_{-\pi}^{\pi} D_m(\xi) f(0) d\xi \\ &= f(0) + \frac{1}{2\pi} \int_{-\pi}^{\pi} (f(\xi) - f(0)) \frac{\sin((m + 1/2)\xi)}{\sin(\xi/2)} d\xi \end{aligned}$$

However, $f'(0)$ exists, so by L'Hopital's rule,

$$\lim_{\xi \rightarrow 0} \frac{f(\xi) - f(0)}{\sin(\xi/2)} = \lim_{\xi \rightarrow 0} \frac{f(\xi) - f(0)}{\xi} \frac{\xi}{\sin(\xi/2)} = 2f'(0)$$

Dirichlet's theorem (1829)

Proof continued. So now we know that this function is continuous (removable discontinuity):

$$h(x) = \begin{cases} \frac{f(x)-f(0)}{\sin(x/2)}, & x \neq 0 \\ 2f'(0), & x = 0 \end{cases}$$

Thus by the Riemann-Lebesgue lemma:

$$\begin{aligned} \lim_{m \rightarrow \infty} \int_{-\pi}^{\pi} D_m(\xi) f(\xi) d\xi &= f(0) + \frac{1}{2\pi} \lim_{m \rightarrow \infty} \int_{-\pi}^{\pi} h(\xi) \sin((m+1/2)\xi) d\xi \\ &= f(0) + 0 = f(0). \end{aligned}$$



Riemann-Lebesgue lemma for continuous functions

Lemma

If $g \in C[a, b]$ then $\lim_{n \rightarrow \infty} \int_a^b g(x) e^{inx} dx = 0$.

Proof. Let $\epsilon > 0$. Since g is continuous on the compact set $[a, b]$, it is uniformly continuous, so there is $\delta > 0$ such that $|x - y| < \delta \implies |g(x) - g(y)| < \epsilon/(2(b - a))$. In particular we can define a mesh of points $\{x_i\}_{i=0}^M$, with spacing less than δ , and x_i^* at the midpoints of each $[x_{i-1}, x_i]$, so that

$$\tilde{g}(x) = \sum_{i=1}^M g(x_i^*) \mathbb{1}_{[x_{i-1}, x_i]}(x)$$

is a uniform approximation of g : $\|\tilde{g} - g\|_\infty = \sup_{x \in [a, b]} |\tilde{g}(x) - g(x)| < \frac{\epsilon}{2(b - a)}$

picture:

Riemann-Lebesgue lemma for continuous functions

Proof continued. Now

$$\begin{aligned} \left| \int_a^b g(x) e^{inx} dx \right| &\leq \left| \int_a^b (g(x) - \tilde{g}(x)) e^{inx} dx \right| + \left| \int_a^b \tilde{g}(x) e^{inx} dx \right| \\ &\leq \|g - \tilde{g}\|_\infty \int_a^b |e^{inx}| dx + \left| \int_a^b \sum_{i=1}^M g(x_i^*) \mathbb{1}_{[x_{i-1}, x_i)}(x) e^{inx} dx \right| \\ &\leq \frac{\epsilon}{2(b-a)}(b-a) + \sum_{i=1}^M |g(x_i^*)| \left| \int_{x_{i-1}}^{x_i} e^{inx} dx \right| \end{aligned}$$

But each of the final integrals is bounded by the same constant:

$$\left| \int_{x_{i-1}}^{x_i} e^{inx} dx \right| = \left| \frac{e^{inx_i} - e^{inx_{i-1}}}{in} \right| \leq \frac{2}{n}$$

Choose N so that $(2\|g\|_\infty M)/N < \epsilon/2$. If $n \geq N$ then

$$\left| \int_a^b g(x) e^{inx} dx \right| \leq \frac{\epsilon}{2} + \frac{2}{n} \sum_{i=1}^M |g(x_i^*)| \leq \frac{\epsilon}{2} + \frac{2}{N} \sum_{i=1}^M \|g\|_\infty < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

□

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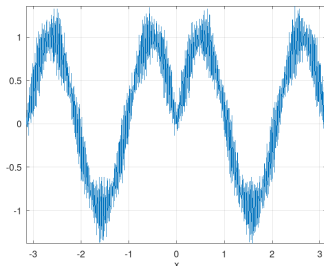
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does the Fourier series converge for any $f \in C[-\pi, \pi]$?

- in 1873 Du Bois-Reymond constructed a continuous function for which the Fourier series **does not** converge at a point

- found by google search:
$$f(x) = \sum_{k=1}^{\infty} \frac{1}{k^2} \sin\left(\frac{(2k^3 + 1)|x|}{2}\right)$$

- the function is even and continuous,¹ so its Fourier series is a cosine series
- it can be shown that the Fourier partial sums are unbounded in a neighborhood of $x = 0$



- in the early 1900s, further examples were constructed where convergence failed on infinite sets of points
- to state the full situation we need to define *measure zero* . . .

¹use Weierstrass M-test

Definition

- a set $E \subset \mathbb{R}$ has (*Lebesgue*) *measure zero* if for any $\epsilon > 0$ there exists a countable list of open intervals $I_n = (a_n, b_n)$ so that

$$E \subset \bigcup_{n \in \mathbb{N}} I_n \quad \text{and} \quad \sum_{n \in \mathbb{N}} m(I_n) = \sum_{n \in \mathbb{N}} b_n - a_n < \epsilon$$

- if a proposition holds except on a set E which has measure zero then we say that it holds *almost everywhere*
- note that any open subset of $\mathbb{R}$ is a countable union of open intervals
- more on Lebesgue measure m , and measure zero, in Chapter 3

the full story for Fourier series of continuous functions

Theorem (Carleson, 1966)

if $f \in C[-\pi, \pi]$, or even $f \in L^p[-\pi, \pi]$, then the Fourier series of f converges pointwise almost everywhere:

$$\lim_{m \rightarrow \infty} \sum_{n=-m}^m c_n e^{inx} = f(x) \quad \text{except on } x \in E \text{ of measure zero}$$

Theorem (Katznelson, 1966)

for any set $E \subset [-\pi, \pi]$ of measure zero there is a continuous periodic function $f \in C[-\pi, \pi]$ whose Fourier series diverges at all points of E

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orthonormal sequences in inner product spaces

- the abstract ideas of Hilbert and Riesz, circa 1910, are fundamentally *simpler* than all of the previous material in these slides
- consider an abstract inner product vector space $V, \langle \cdot, \cdot \rangle$

Definition

- a pair of vectors $v, w \in V$ are *orthogonal* if $\langle v, w \rangle = 0$
- a set S is *orthogonal* if $v, w \in S$ and $v \neq w$ implies v, w are orthogonal
- $(v_k)_{k=1}^{\infty} \subset V$ is an *orthonormal sequence* (*ON sequence*) if v_k, v_l are orthogonal for $k \neq l$ and if $\|v_k\|^2 = \langle v_k, v_k \rangle = 1$

Lemma

- *an orthogonal set is linearly-independent*
- *an ON sequence is linearly-independent*

examples of orthonormal sequences

- $\left\{ \sqrt{\frac{2}{\pi}} \sin(nx) \right\}_{n \in \mathbb{N}}$ is a (real) ON sequence on $[0, \pi]$
- $\left\{ \frac{1}{\sqrt{2\pi}} e^{inx} \right\}_{n \in \mathbb{Z}}$ is an ON sequence on $[-\pi, \pi]$
- $\left\{ \frac{1}{\sqrt{2\pi}} e^{inx} \right\}_{n \in \mathbb{N}}$ is an ON sequence on $[-\pi, \pi]$
- $\left\{ \sqrt{\frac{2}{\pi}} \cos(nx) \right\}_{n \in \mathbb{N}}$ is a (real) ON sequence on $[0, \pi]$
- $\left\{ \sqrt{\frac{1}{\pi}} \sin(nx) \right\}_{n \in \mathbb{N}}$ is a (real) ON sequence on $[-\pi, \pi]$
- the Legendre polynomials² form an ON sequence

²with a different normalization these are: $1, x, \frac{1}{2}(3x^2 - 1), \frac{1}{2}(5x^3 - 3x), \dots$

examples of orthonormal sequences

- $\left\{ \sqrt{\frac{2}{\pi}} \sin(nx) \right\}_{n \in \mathbb{N}}$ is a (real) ON sequence on $[0, \pi]$ $\leftarrow$ complete
- $\left\{ \frac{1}{\sqrt{2\pi}} e^{inx} \right\}_{n \in \mathbb{Z}}$ is an ON sequence on $[-\pi, \pi]$ $\leftarrow$ complete
- $\left\{ \frac{1}{\sqrt{2\pi}} e^{inx} \right\}_{n \in \mathbb{N}}$ is an ON sequence on $[-\pi, \pi]$ $\leftarrow$ not complete
- $\left\{ \sqrt{\frac{2}{\pi}} \cos(nx) \right\}_{n \in \mathbb{N}}$ is a (real) ON sequence on $[0, \pi]$ $\leftarrow$ not complete
- $\left\{ \sqrt{\frac{1}{\pi}} \sin(nx) \right\}_{n \in \mathbb{N}}$ is a (real) ON sequence on $[-\pi, \pi]$ $\leftarrow$ not complete
- the Legendre polynomials² form an ON sequence $\leftarrow$ complete

²with a different normalization these are: $1, x, \frac{1}{2}(3x^2 - 1), \frac{1}{2}(5x^3 - 3x), \dots$

fundamental question about an orthonormal sequence

question

- given an ON sequence $S = (f_k)$ from an inner product space $(V, \langle \cdot, \cdot \rangle)$, is its span dense in V ?

$$\text{is } \overline{\text{span } S} = V?$$

- this question becomes a definition ...

Definition

an ON sequence $S = (f_k)$ in an inner product space $(V, \langle \cdot, \cdot \rangle)$ is *complete* if for every $f \in V$ there exist coefficients c_k so that

$$f = \lim_{N \rightarrow \infty} \sum_{k=1}^N c_k f_k = \sum_{k=1}^{\infty} c_k f_k$$

- language: *complete ON sequence* = *ON basis*

Theorem 4.1

Suppose that $f = \sum_{k=1}^{\infty} c_k f_k$ for an ON sequence $S = (f_k)$ in an inner product space $V, \langle \cdot, \cdot \rangle$. Then

$$c_k = \langle f, f_k \rangle$$

- the proof is easy . . . we will do it

Definition

- for $f \in V$, the *Fourier coefficients* for an ON sequence $S = (f_k)$ are

$$c_k = \langle f, f_k \rangle$$

- the *Fourier series* of $f \in V$, for an ON sequence $S = (f_k)$, is

$$\sum_{k=1}^{\infty} \langle f, f_k \rangle f_k$$

how to prove an ON sequence is complete?

- the Fourier series is always in V , but that does *not* imply, by itself, that it equals f
- if the ON sequence is complete then f equals its Fourier series
- part (c) of the following theorem is the most practical way to show an ON sequence of functions on an interval is complete

Theorem 4.5

for an ON sequence $S = (f_k)_{k \in \mathbb{N}}$ in $L^2 = L^2([-\pi, \pi], m)$, the following are equivalent:

- (a) S is a complete ON sequence
- (b) for every $f \in L^2$ and $\epsilon > 0$ there is a finite linear combination $g = \sum_{k=1}^n d_k f_k$ so that $\|f - g\|_2 < \epsilon$
- (c) if the Fourier coefficients of $f \in L^2$, for the ON sequence S , are zero then the function f is zero almost everywhere

- we will prove this Theorem

Fourier's claim is true *if* convergence is in L^2

Theorem 4.6 (rewritten in complex form)

The ON sequence $S = \left(\frac{1}{\sqrt{2\pi}} e^{inx} \right)_{n \in \mathbb{Z}}$ is complete in $L^2([-\pi, \pi], m)$. Therefore

$$f(x) = \frac{1}{\sqrt{2\pi}} \sum_{n=-\infty}^{\infty} c_n e^{inx},$$

for these coefficients

$$c_n = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} f(x) e^{-inx} dx,$$

in the sense that **the partial sums converge to f in the norm $\|\cdot\|_2$**

- this is the simplest interpretation of Fourier's claim
 - however, it does address continuous functions directly
- we will get to L^2 spaces seriously in Chapter 3

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theory in weeks 4 & 5: Chapter 4 in Saxe

- we will do Fourier series (Chapter 4) *before* Lebesgue integrals and measures (Chapter 3)

to define:

- $L^2([a, b], m)$ and $L^2(X, \mu)$ ← not fully defined till Chapter 3
- complete orthonormal sequence
- Fourier series and coefficients for any orthonormal sequence

to prove:

- Bessel's inequality
 - Parseval's equality
 - the Riesz-Fischer theorem (in old and “full-power” forms)
 - Theorem 4.6: Fourier series converge in L^2
-
- Assignment 3 is posted at bueler.github.io/fa